

DFR-FORENSICS

Disk & File Resurrection : Low-Level Architecture & Technical Whitepaper

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1. Forensic Problem Statement & Architectural Philosophy

In modern cyberattacks (destructive wipers like *HermeticWiper*, *CaddyWiper*, *WhisperGate*) and catastrophic storage hardware incidents (initial bad sector bursts, power loss during partitioning), storage boot structures are systematically targeted. When sector LBA 0 (MBR) or LBA 1 (GPT) is wiped with zeroes, operating systems (Windows, Linux, macOS) and traditional forensic tools flag the media as **completely unallocated or uninitialized**.

DFR-Forensics (*Disk & File Resurrection*) was designed by **Dam-FOR3K** to bridge this critical gap through strict forensic paradigms:

- **Strict Zero-Write Policy:** No write operations are ever performed on the target evidence. Raw images (RAW, DD, E01, AD1, AFF4) and live physical disks (`\\.\PhysicalDriveX`) are opened with strict read-only hardware flags.
- **Virtual Copy-On-Write (COW) Architecture:** All table restorations, container decryptions, and superblock syntheses are executed purely in an in-memory virtual layer, ensuring 100% evidence integrity.
- **Mathematical Reconstruction via Invariants:** No offsets or cluster sizes are hardcoded. The engine derives architecture from universal filesystem laws (media descriptors, cluster chain relationships, and magic headers).

2. Low-Level Partition Table Architecture

A. Master Boot Record (MBR) & EBR Chains

Sector LBA 0 hosts the legacy MBR: 446 bytes of bootstrap code, 4 partition descriptor entries of 16 bytes each (offsets 446 to 509), and the boot signature 0x55AA (offsets 510-511). Each entry stores the boot flag, filesystem type ID (e.g. 0x07 NTFS/exFAT, 0x83 Linux Native), starting LBA, and total sector count. When extended partitions are used, the tool recursively navigates the linked chain of **Extended Boot Records (EBR)**.

B. UEFI GPT Specification (GUID Partition Table 2.10)

Modern storage adheres to the UEFI GPT layout:

- **LBA 0 (Protective MBR) :** Dummy MBR containing a single partition of type 0xEE spanning the entire drive to prevent legacy utilities from misidentifying the disk.
- **LBA 1 (Primary GPT Header) :** 92-byte header beginning with signature EFI PART (0x5452415020494645). Stores disk GUIDs, partition table pointer (LBA 2), backup header pointer (LBA N-1), and two independent CRC32 checksums.
- **LBA 2 to 33 (Partition Entry Array) :** 128 entries of 128 bytes each (16,384 bytes). Each record specifies partition type GUID, unique GUID, first LBA, last LBA, attributes, and UTF-16LE partition label.
- **LBA N-33 to N-2 (Backup Array) and LBA N-1 (Backup Header) :** Full replica at the end of the disk providing fault tolerance.

C. CRC32 Recalculation Algorithm (IEEE 802.3)

When a wiper zeroes out the first megabytes of a disk (LBA 0..2048), the Primary GPT is wiped, but the Secondary GPT (LBA N-1) is frequently intact. DFR-Forensics restores the structure as follows :

1. Extract partition array from LBA N-33 and map to LBA 2..33.
2. Invert geometry pointers: $\text{CurrentLBA} = 1$, $\text{BackupLBA} = \text{N}-1$, $\text{PartitionEntriesLBA} = 2$.
3. Recalculate CRC32 of partition array (16,384 bytes) using reflected polynomial 0xEDB88320.
4. Reset the 4 bytes of Header CRC32 (offset 16) to zero, calculate CRC32 over the 92-byte header, and inject back.
5. Synthesize a valid Protective MBR at LBA 0.

3. Filesystem Engines & Autonomous Recovery Algorithms

A. FAT12 / FAT16 / FAT32 & Backup Boot Sector Failover

In FAT filesystems, the **BPB (BIOS Parameter Block)** in the Volume Boot Record (LBA 0) dictates all cluster geometry: bytes per sector, reserved sectors, number of FAT tables, root entries, and sectors per cluster (SPC). When LBA 0 is zeroed out by a wiper, **DFR-Forensics** applies a dual-tier resilience strategy:

- **Backup Boot Sector Failover (LBA 6):** Under FAT32, the engine probes sector 6 (official OEM boot sector backup). If intact, original volume geometry is restored instantly without heuristic derivation.
- **Mathematical Orphan BPB Reconstruction:** When both sectors 0 and 6 are destroyed:
 1. Scans media descriptor bytes (0xF8 / 0xF0) to locate FAT1.
 2. Computes: **Sectors Per FAT = LBA(FAT2) - LBA(FAT1)**.
 3. Positions the Root Directory table immediately following FAT2.
 4. Dynamically solves cluster size (SPC) by testing candidate offsets against real file signatures (JPEG, PDF, ZIP).

B. Linux EXT2 / EXT3 / EXT4 & Extents Tree Architecture

Linux EXT organizes storage into Block Groups. When the primary superblock is wiped (LBA 2) or files are fragmented:

- **Sparse Superblock Discovery:** Automatically sweeps offset 8,389,632 (LBA 16,386) and group boundaries up to 32 MB to detect magic 0xEF53 and restore group descriptors.
- **Full EXT4 Extents Support (magic 0xF30A):** Native traversal of the extent tree stored within modern 60-byte inode blocks. Recursively decodes index nodes (depth > 0) and leaf extent descriptors (ee_block, ee_start, ee_len), ensuring bit-exact reassembly of multi-gigabyte fragmented files.
- **Direct, Indirect & Double Indirect Pointer Resolution:** Chirurgical traversal of legacy Ext2/3 block trees, skipping 1,024-byte pointer metadata blocks interspersed throughout large files.
- **Directory Slack Space Undelete:** Unlinked file records absorbed by active entries' rec_len are extracted with original names and metadata.

C. Windows NTFS (\$MFT, \$MFTMirr & Backup VBR)

Direct pure-Python **\$MFT** parsing with critical failover mechanisms:

- **\$MFTMirr Failover (cluster offset 0x38):** When Record 0 of the primary \$MFT is damaged or wiped, the engine fails over to \$MFTMirr to recover non-resident \$DATA runlists mapping the entire MFT across disk clusters.
- **Backup VBR Resolution (Last Sector LBA N-1):** When the partition boot sector is blank, BPB parameters (cluster size, record size, MFT cluster) are retrieved from the volume end.
- **Transaction Log Carving (\$LogFile) & Undelete:** Comprehensive reconstruction of active and deleted files with quadruple MACB timestamps (\$STANDARD_INFORMATION).

D. Native exFAT Engine (Main & Backup VBR)

Pure-Python forensic parser for flash and external storage media: Backup VBR failover (sector 12), directory entry chain parsing (primary active 0x85 / deleted 0x05, stream extension 0xC0 with contiguous allocation flag NoFatChain, and Unicode name sequences 0xC1). Direct bit-exact extraction without third-party drivers.

E. Embedded & Unix Systems (QNX, APFS, LUKS, BitLocker)

- **QNX4 & QNX6 Power-Safe:** Automotive head units and embedded controllers via superblocks 0x68191122 at LBA 8192/11776 and transaction logs.
- **Apple APFS:** Container Superblock NXSB, Object Map (OMAP) B-Tree traversal, and multi-volume container enumeration.
- **BitLocker & LUKS1/2:** Metadata extraction, PBKDF2/Argon2id key derivation, and transparent in-memory streaming decryption.

4. Intelligent Carving Engine & Resilient Preview Cocktail

Unlike naive carvers that extract arbitrary chunks and flood analysts with false positives, **DFR-Forensics** enforces format-specific structural and mathematical validation:

Format	Header Signature	Validation Algorithm & Length Computation
JPEG / JFIF	FF D8 FF E0..FE	Sequential marker parsing (SOF, DQT, DHT, SOS). MCU block validation and strict search for EOI marker FF D9.
PNG	89 50 4E 47 0D 0A 1A 0A	Full CRC-32 (IEEE 802.3) integrity verification on each chunk (IHDR, IDAT, IEND). File boundary locked at the final IEND chunk byte.
BMP	42 4D ('BM')	Little-endian total filesize verification from bytes 2..5. Non-negative dimensions and valid DIB header validation.
GIF	GIF87a / GIF89a	Logical Screen Descriptor dimensions check, full block structure traversal (extensions 0x21, image descriptors 0x2C, LZW sub-blocks) down to the legitimate trailer 0x3B for byte-exact boundary determination.
OLE CFBF (DOC, XLS, PPT)	D0 CF 11 E0 A1 B1 1A E1	Internal OLE FAT traversal to compute exact sector allocation and avoid truncated documents.
PDF	%PDF-1.x	Bounded forward scan for the nearest %%EOF marker to prevent document concatenation.
TIFF 6.0	49 49 2A 00 ('II*\x00') 4D 4D 00 2A ('MM\x00*')	Full Little-Endian and Big-Endian support. Traverses Image File Directory (IFD), computes exact physical span via Strip/Tile ByteCounts, and filters embedded JPEG EXIF false positives.
ZIP / Office XML	50 4B 03 04 ('PK..')	Local File Header chaining and validation via Central Directory (PK\x01\x02) and EOCD (PK\x05\x06).

Strategic Carving Options & Sector Alignment

To adapt to any storage topology or memory dump, the engine offers surgical configuration options:

- **512 bytes (Standard sectors - Recommended):** High-speed default mode aligned on physical sector boundaries. Ideal for HDD, SSD, USB flash drives, and SD cards.
- **1 byte (Exhaustive / Any offset):** Byte-by-byte exhaustive sweep (unaligned). Indispensable for RAM dumps, network streams, or shifted raw disk images where files begin at arbitrary offsets (1, 2, 17 bytes...).
- **4,096 bytes (Standard clusters):** Accelerated scan aligned on standard filesystem cluster boundaries (NTFS, ext4).
- **Unallocated Space Only (Deleted Files):** Targets exclusively freed and unassigned disk clusters by interrogating filesystem allocation tables and bitmaps (NTFS \$Bitmap, FAT tables, exFAT allocation bitmap, EXT2/3/4 block bitmaps, QNX4/6 allocation tables, Apple APFS Spaceman & containers, and unpartitioned gaps). This strategy eliminates duplicate extraction of healthy active files and accelerates the carving scan by 5x to 10x.
- **Advanced De-Braiding (BraidResolver):** Disabled by default to preserve raw file integrity and prevent accidental fragmentation on standard systems. When explicitly enabled by the analyst, it mathematically separates interleaved file pairs (1A-1B-2A-2B pattern) with verified in-memory pixel decompression.

NIST CFTT Graphic Carving Compliance & Automated De-Braiding

DFR-Forensics successfully passes the complete official **NIST CFTT Graphic Carving** test suite (8 forensic benchmark disk images):

- **graphic-basic & graphic-nofill:** 40/40 contiguous files (BMP, GIF, JPG, PNG, TIFF) carved with 100% bit-exact dimensions and hashes.
- **graphic-shifted & notshifted:** Lossless extraction of unaligned streams (offsets 1 to 511 bytes within sectors).
- **graphic-partials:** Resilient recovery of truncated image streams without parser failures.

- **graphic-braid (In-Memory De-Braiding):** The BraidResolver engine automatically identifies and mathematically disentangles interleaved file pairs [1A, 1B, 2A, 2B] without Frankenstein pixel corruption (100% of 20 images successfully decoded in memory).

The Resilient Preview Cocktail for Damaged Images

Real-world forensic images often exhibit partial corruptions. The tool integrates a three-tier resilience cocktail:

1. **Truncated Stream Auto-Closing:** Injects virtual FF D9 in memory if EOI is missing, allowing graphical renderers to display all intact MCUs up to the cut.
2. **Permissive Decoding Mode:** Pillow fallback with LOAD_TRUNCATED_IMAGES = True when strict Qt parsers reject damaged images.
3. **Surgical Header Tolerance:** Dynamically patches corrupted marker lengths (e.g. corrupted DQT table length FF DB 00 00) in memory to achieve 100% visual preview.

5. Multi-Tier Shannon Entropy & Forensic Resilience Validation

Four-Tier Forensic Entropy Scale

Shannon entropy measures information randomness: $H(X) = -\sum(p(x) * \log_2(p(x)))$. To avoid misidentifying normal compressed media as encrypted storage, the disk presence bar utilizes a 4-tier scale:

- **Emerald Green (Clear Active Data - $H \leq 7.4$):** Plain text, partition tables, metadata, and uncompressed structures.
- **Dodger Blue / Cyan (Compressed & Media - $7.4 < H \leq 7.88$):** JPEG, MP4/WMV/MOV, ZIP archives, and PDFs.
- **Deep Purple (High Entropy / True Encryption - $H > 7.88$):** BitLocker, LUKS, and cryptographic keystores.
- **Bordeaux Black (100% Zeroes / Wiped Space - Ratio > 98%):** Wiped or unallocated sectors.

Corruption Scenarios & Resilience Validation Matrix

Corruption Scenario	Target System	Applied Corruption	Recovery Engine Mechanism	Data Integrity
Severed Boot Sector (VBR)	FAT16 / FAT32	Boot Sector LBA 0 wiped (zeroes) Missing partition table	Autonomous BPB derivation via mirror FAT & cluster magic correlation	100% Bit-Exact Files & directory tree restored
Severed Primary Superblock	Linux EXT2/3/4	Primary superblock wiped Deleted files in directory slack space	Backup superblock failover, double-indirect defragmentation & undelete	100% Bit-Exact Full tree & hashes verified
Steganographic Volume Label	FAT12/16/32	File payload concealed under Volume Label attribute (0x08)	Directory metadata attribute analysis & automated payload extraction	100% Intact Stego anomaly flagged & dumped
Fragmented Extended Partitions	DOS MBR / Extended	Complex nested EBR linked chain with unallocated gaps	Recursive EBR parsing & orphan boot record discovery	100% Reconstructed All logical drives recovered
Truncated & Corrupted Media	JPEG / PNG / BMP / OLE / PDF	Missing EOF marker, header table byte corruption	Resilience cocktail: auto-closure (FF D9), header tolerance & permissive decoding	Visual Recovery Preview restored seamlessly

Conclusion

By fusing low-level UEFI GPT 2.10 repair, autonomous orphan BPB derivation, EXT2/3/4 defragmentation, and granular intelligent carving, **DFR-Forensics** stands as a definitive, battle-tested forensic instrument for law enforcement agencies, digital forensics labs, CSIRT/SOC incident responders, and data recovery specialists.